Performance and symmetry measures during vertical jump testing at return to sport after ACL reconstruction

Roula Kotsifaki ,[1,2] Vasileios Sideris,[1] Enda King ,[1,3] Roald Bahr ,[2,4] Rod Whiteley [1,5]

► Additional supplemental material is published online only. To view, please visit the journal online (http://dx.doi.org/10.1136/bjsports-2022-106588).

[1]Rehabilitation Department, Aspetar Orthopaedic and Sports Medicine Hospital, Doha, Qatar
[2]Oslo Sports Trauma Research Center, Department of Sports Medicine, Norwegian School of Sport Sciences, Oslo, Norway
[3]Department of Life Sciences, Roehampton University, London, UK
[4]Aspetar Sports Injury and Illness Prevention Programme (ASPREV), Aspetar, Orthopeadic and Sports Medicine Hospital, Doha, Qatar
[5]School of Human Movement & Nutrition Sciences, The University of Queensland, Saint Lucia, Queensland, Australia

Correspondence to
Dr Roula Kotsifaki, Rehabilitation Department, Aspetar Orthopaedic and Sports Medicine Hospital, Doha P.O. Box 29222, Qatar; argyro.kotsifaki@aspetar.com

Accepted 6 May 2023

Check for updates

© Author(s) (or their employer(s)) 2023. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: Kotsifaki R, Sideris V, King E, *et al*. *Br J Sports Med* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/bjsports-2022-106588

ABSTRACT

Objective Vertical jump tests are more sensitive in revealing asymmetries in performance metrics at the time of return to sport after anterior cruciate ligament (ACL) reconstruction (ACLR) than horizontal hop tests. However, it remains unclear which vertical tests (bilateral or unilateral) and which metrics (kinetics or performance) are most effective in informing the rehabilitation status and readiness for return to sport. We aimed to investigate the status of athletes during vertical jump testing at return to sport after ACLR.

Methods A dual force platform system was used to evaluate jumping performance of 126 recreational and professional athletes at the time of return to sport after ACLR, as well as 532 healthy control participants. Performance and kinetic metrics were collected during four jump tests: double-leg countermovement jump, single-leg countermovement jump, double-leg 30 cm drop jump and single-leg 15 cm drop jump. Between-limb and between-group differences were explored using mixed models analyses.

Results At the time of return to sport after ACLR, athletes still presented significant differences favouring the uninvolved side, particularly in the symmetry of the concentric impulse (p<0.001) in all jumps compared with the control group. Peak landing force asymmetry was greater in the ACLR group than the controls during the countermovement (p<0.001, MD=−11.6; 95% CI −15.4 to −7.9) and the double-leg drop jump (p=0.023, MD=−8.9; 95% CI −14.9 to −2.8). The eccentric impulse asymmetry was significantly greater (p=0.018, MD=−3.8; 95% CI −5.8 to −1.7) in the ACLR group during the single-leg drop jump only. Jump height was significantly lower (p<0001) in the ACLR group compared with controls in all tests except the double-leg drop jump.

Conclusion At the time of return to sport after ACLR, despite passing the traditional discharge criteria, athletes remained asymmetrical during all vertical jump tests, in the concentric (push-off) phase, during landing from bilateral jumps and for most performance metrics. Clinicians should aim to restore not only symmetry in ground reaction forces but also absolute performance metrics such as jump height, reactive strength index and contact times, to potentially reduce injury risk and improve overall athletic performance.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ At the time of return to sport, athletes still present biomechanical asymmetries (joint angles, moments and work and muscle forces) during vertical jumps, despite passing discharge criteria.

⇒ Ground reaction force analysis during vertical jump testing can offer valuable phase-specific information on the status of an athlete after anterior cruciate ligament reconstruction (ACLR).

WHAT THIS STUDY ADDS

⇒ We report the metrics that are not yet restored at the time of return to sport, and therefore, likely to be of clinical interest.

⇒ Asymmetries were present in the concentric phase (push-off), during landing and in most of the performance metrics.

⇒ We provide normative data of performance metrics during vertical jumps of a large cohort of professional football players.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Vertical jumps (single and bilateral) should be included in the periodic and return to sport testing after ACLR.

⇒ Clinicians should monitor and restore symmetry and absolute performance metrics such as jump height, reactive strength index and contact times.

⇒ Plyometric training should be an integral part of the rehabilitation protocol given the observed residual deficits seen in the single-leg drop jump test.

INTRODUCTION

Functional hop testing is traditionally used to determine readiness to return to sport (RTS) after anterior cruciate ligament reconstruction (ACLR)[1] and horizontal hops are the most commonly reported tests.[2] Recent studies highlighted that horizontal hop distance is not a sensitive metric to evaluate readiness to RTS.[3–5] Symmetry in performance on a single or triple hop for distance does not ensure symmetry in lower limb biomechanics such as: joint angles, moments and power, and muscle force contribution.[4 5] However, the hop for distance task, specifically the landing phase, can offer valuable information on the status of the knee joint—the joint-specific contributions and possible intralimb compensations from the hip and ankle joints.[5] However, such assessments require three-dimensional (3D) biomechanical analysis, apparatus not frequently available in the clinical setting.

In contrast, vertical jump performance testing is sensitive in identifying between limb-asymmetries

Br J Sports Med: first published as 10.1136/bjsports-2022-106588 on 1 June 2023. Downloaded from http://bjsm.bmj.com/ on June 5, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.

at the end phase of rehabilitation after ACLR[6 7] with symmetry in vertical jump height more difficult to achieve than symmetry in horizontal hop distance.[8–11] In healthy athletes, vertical jump testing is a common performance assessment due to its relative simplicity and time efficiency.[12–14]

In most sports and clinical environments, the use of 3D motion capture and triaxial force plates for biomechanical assessment is not possible due to financial and logistic constraints. However, the use of dual force platform single axis technology is increasingly common, allowing the assessment of vertical ground reaction forces and asymmetries during double and single-leg jump activities.[15] Furthermore, affordable solutions such as contact measuring devices, mobile apps etc can be used to accurately assess jump performance.[16–21]

Objective testing is an integral part of the rehabilitation and the shared decision-making process to allow an athlete to RTS.[22] However, it is as yet unclear which tests and which metrics clinicians should use to track the progress of athletes during rehabilitation after ACLR, especially at the time of RTS. There are a variety of vertical tests available (countermovement and drop jumps, double and single leg) and hundreds of metrics routinely provided by the software platforms for these tests, an overwhelming data volume for the clinician unsure which are relevant.[14]

Our hypothesis was that athletes after ACLR would still display between-limb differences as well as compared with a healthy control group during the concentric, eccentric and landing phases during four single-leg and double-leg vertical tests, despite being cleared to return to sport. Accordingly, our goal was to explore the status of the ACLR athletes for these metrics at the time of RTS. We aimed to identify which metrics still displayed deficits using only force plates with their manufacturer-provided auto-detection software.

Methods

We included 658 male participants, 126 eligible patients after ACLR and 532 control subjects tested in the assessment unit at Aspetar Orthopaedic and Sports Medicine Hospital, Doha, Qatar from July 2017 to July 2022 (table 1). Patients with ACLR were enrolled after the completion of a standardised rehabilitation protocol and after receiving clearance to RTS. The criteria for RTS were: (1) clearance by both their surgeon and physiotherapist, (2) completion of a sports-specific on-field rehabilitation programme, (3) quadriceps strength Limb Symmetry Index (LSI) >90% and (4) hop battery tests with LSI>90%.[1] ACLR patients were professional or recreational athletes with a complete, unilateral ACL injury, reconstructed either with an autologous ipsilateral bone-patellar-tendon-bone (BTB) or a hamstring graft (HS) (semitendinosus and gracilis), as clinically decided by the surgeon and athlete. Participants were excluded if they had concomitant grade III knee ligament injury (other than ACL). Control subjects were male professional athletes recruited from a cohort of professional soccer players as part of an annual periodic health evaluation at the same venue. Inclusion criteria for the control group were: age range of 18–35 years, football players with no previous surgery and no history of musculoskeletal injury of the lower limb during the 3 months prior to testing. All participants provided informed consent, and the study was approved by the local institutional review board (F2017000227 and E202009010).

Table 1 Participant information

	ACLR group (n=126)	Control group (n=532)
Professional athletes (n)	94	532
Age (years)	24±6	25±5
Body mass (kg)	75±14	74±10
Height (cm)	176±9	177±7
Tegner score preinjury	8.5±1.2 (range 5–10)	9.3±0.4 (range 9–10)
Return to sport (months)	9.4±2.6	NA
ACL graft type, n (%)		
Hamstring	72 (57)	
Bone-patellar-tendon-bone	54 (43)	
Isolated ACL injury	75 (60)	
Meniscal surgery	51 (40)	
Meniscectomy	17 (13)	
Meniscus repair	34 (27)	
Lateral augmentation	62 (49)	
Lateral tenodesis	42 (33)	
Anterolateral ligament	20 (16)	

All participants were male. Values other than number of participants are expressed as mean±SD.
ACLR, anterior cruciate ligament reconstruction; NA, not available.

Equity, diversity and inclusion statement

Our study was on male athletes after ACLR and controls and no potential participant was excluded based on race/ethnicities, socioeconomic levels and marginalised groups. We did not include females due to the small number of female athletes tested (4) who met the inclusion criteria in our institution. Our author team consisted of one female (first author) and four male, junior, mid-career, and senior researchers from different disciplines, and different ethnicities.

PROCEDURES

The complete battery of tests included, in order, the following: double countermovement jump, single-leg countermovement jump, double-leg drop jump from 30 cm box and single-leg drop jump from a 15 cm box. Athletes were instructed to keep their hands on their hips and jump maximally.

The jumps were examined using two ground-embedded force plates (AMTI, Watertown, Massachusetts, USA) or a dual force plate system (ForceDecks, Vald Performance, Newstead, Australia) at a sampling rate of 1000 Hz. Data were recorded with ForceDecks software (Vald Performance, Newstead, Australia) for both hardware configurations and were analysed with ForceDecks software V.2.0. All participants followed a standardised warm-up protocol that included 5 min cycling on static bike and three double-leg countermovement jumps at submaximum effort (perceived intensity 50%, 75% and 90%), followed by two task-specific jumps. After zeroing the plates, and weighing the athlete, each test was first demonstrated to, and then practised by the participant. The ForceDecks software automatically identifies jump types, along with a range of metrics. The calculation methods by the software for all outcomes of interest are described in detail in online supplemental file 1. Events were identified by a 20 N change in ground reaction force. The jump-type identification was verified by the operator, along with jump quality, with manual removal of any misidentified or poorly performed jumps (eg, landing outside the force plates, arm swing).

Countermovement jump

Athletes were instructed to stand fully upright, keep their hands on hips and remain motionless for a minimum of 3 s before the initiation of the test. Athletes were asked to countermove quickly and then jump as high as possible.

Br J Sports Med: first published as 10.1136/bjsports-2022-106588 on 1 June 2023. Downloaded from http://bjsm.bmj.com/ on June 5, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.

Single-leg countermovement jump

Athletes were instructed to stand fully upright, with one foot on the force plate and the free leg behind at approximately 60° knee flexion, keep their hands on hips and remain motionless for a minimum of 3 s before the initiation of the test. Athletes were asked to countermove quickly and then jump as high as possible.

Double-leg drop jump

Athletes were asked to keep their hands on hips, roll from a 30 cm box with both feet together and on hitting the ground, immediately jump as high as possible while spending as little time as possible on the force plate ('as fast and as high as possible').

Single-leg drop jump

Athletes were asked to keep their hands on hips, hold their non-testing leg behind at approximately 60° knee flexion, roll from a 15 cm box and on hitting the ground, immediately jump as high as possible while spending as little time as possible on the force plate ('as fast and as high as possible').

Test limb order was randomised for the control group. For athletes after ACLR, we first tested the uninvolved leg. Limb dominance was determined by asking the participants with which limb they would prefer to kick a ball.[23] Three trials of each test were performed with a 30 s rest period between jumps. The mean value of the three jumps was recorded and used for subsequent analysis. Kinetic outcomes of interest were eccentric impulse, force at zero velocity, concentric impulse and peak landing force. For the performance metrics, variables were extracted from the best jump out of three (in terms of jump height). We used the jump height calculated by the impulse–momentum method and Reactive Strength Index (RSI) (jump height/contact time).[24 25] LSI was determined as the percentage of the involved divided by the uninvolved leg for the ACLR group and non-dominant leg divided by dominant leg for controls for all performance and ground reaction force variables.[2 26] A symmetry of <100% indicates favouring the uninvolved leg for the ACLR group and the non-dominant for the control group.

STATISTICAL ANALYSIS

Descriptive statistics were used to summarise the characteristics of the participants and measurements. Variables representing different aspects (concentric impulse, eccentric impulse, peak force, performance) of each phase of the jump were retained for further analyses provided they displayed sufficient precision (variance estimate compared with the group estimator). Normality of distribution of data was assessed with the Shapiro-Wilk test[27] and Q-Q plots. Between-groups (ACLR and control) comparisons were assessed using mixed effect models with subject-specific random effects. We first examined for any effect of age or activity level on symmetry (of kinetics) and found no differences, so professional and recreational athletes were explored in the same group. We then performed subgroup analyses to evaluate the effect of graft on kinetic symmetry and performance outcomes using one-way analysis of variance with post hoc comparisons adjusted for multiple comparisons (Tukey). For the performance metrics, however, there were significant differences between professional and recreational athletes. Accordingly, for the analyses of performance, comparisons were performed only between professional athletes after ACLR and a control group of professional athletes. For the performance metrics during single-leg jumps in the control group, exploratory analyses examined for differences between dominant/non-dominant legs. As there were no significant differences for these metrics, a randomly chosen leg was used for subsequent analyses. Post hoc comparisons (Tukey) were performed to adjust for multiple comparisons where applicable (single-leg jumps comparing the involved leg, the uninvolved leg and the controls). A $p<0.05$ was considered statistically significant. Effect sizes were calculated using the pooled weighted SD.[28] Analyses were performed using SPSS V.26 (IBMY) and JMP V.16 (SAS Institute).

Results

Participants were tested within 2 weeks of clearance to RTS. Subgroup comparisons of graft types showed some differences for kinetics symmetry (online supplemental file 2). There was an effect of activity level on performance metrics; hence all comparisons on performance outcomes were performed only between professional athletes after ACLR and a control group of professional athletes (online supplemental file 3). We did not include recreational athletes in the comparisons of performance metrics, however, we report their performance metrics for reference. All metrics that were examined are presented in online supplemental file 4.

Countermovement jump

There was no difference between athletes after ACLR and the control group for the symmetry during the eccentric phase, or at the transition between eccentric and concentric (force at zero velocity). Conversely, there were significant differences during the concentric phase, especially in the second half (concentric impulse P2). Peak landing force symmetry was significantly different between groups. Jump height in the professional athletes after ACLR did not reach the values of the control group (figure 1, table 2, online supplemental file 3).

Single-leg countermovement jump

There was no difference between groups for the eccentric deceleration impulse symmetry and no difference for the force at zero velocity symmetry. Similar to the countermovement jump, significant differences were found for the concentric impulse symmetry. There was no difference in peak landing force symmetry. Jump height in the professional athletes after ACLR was significantly less than their uninvolved limb and did not reach the jump height of the control group (figure 1, table 2, online supplemental file 3).

Double-leg drop jump

There was no difference between ACLR and healthy groups in the eccentric impulse symmetry. There was a difference for the force at zero velocity symmetry between groups. Significant differences were observed for the concentric impulse and peak landing force symmetry. In terms of performance, significant differences were found for contact time and RSI but not for jump height between professional athletes after ACLR and the control group (figure 1, table 2, online supplemental file 3).

Single-leg drop jump

There were significant differences in the eccentric impulse, force at zero velocity and the concentric impulse symmetries between athletes after ACLR and the control group. There was no difference in symmetry of peak force during landing. In terms of performance, there were still significant differences between legs in athletes after ACLR for jump height and RSI. Importantly, neither involved nor uninvolved limbs reached the performance values of the control group in any of the performance metrics (figure 1, table 2, online supplemental file 3).

Br J Sports Med: first published as 10.1136/bjsports-2022-106588 on 1 June 2023. Downloaded from http://bjsm.bmj.com/ on June 5, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.

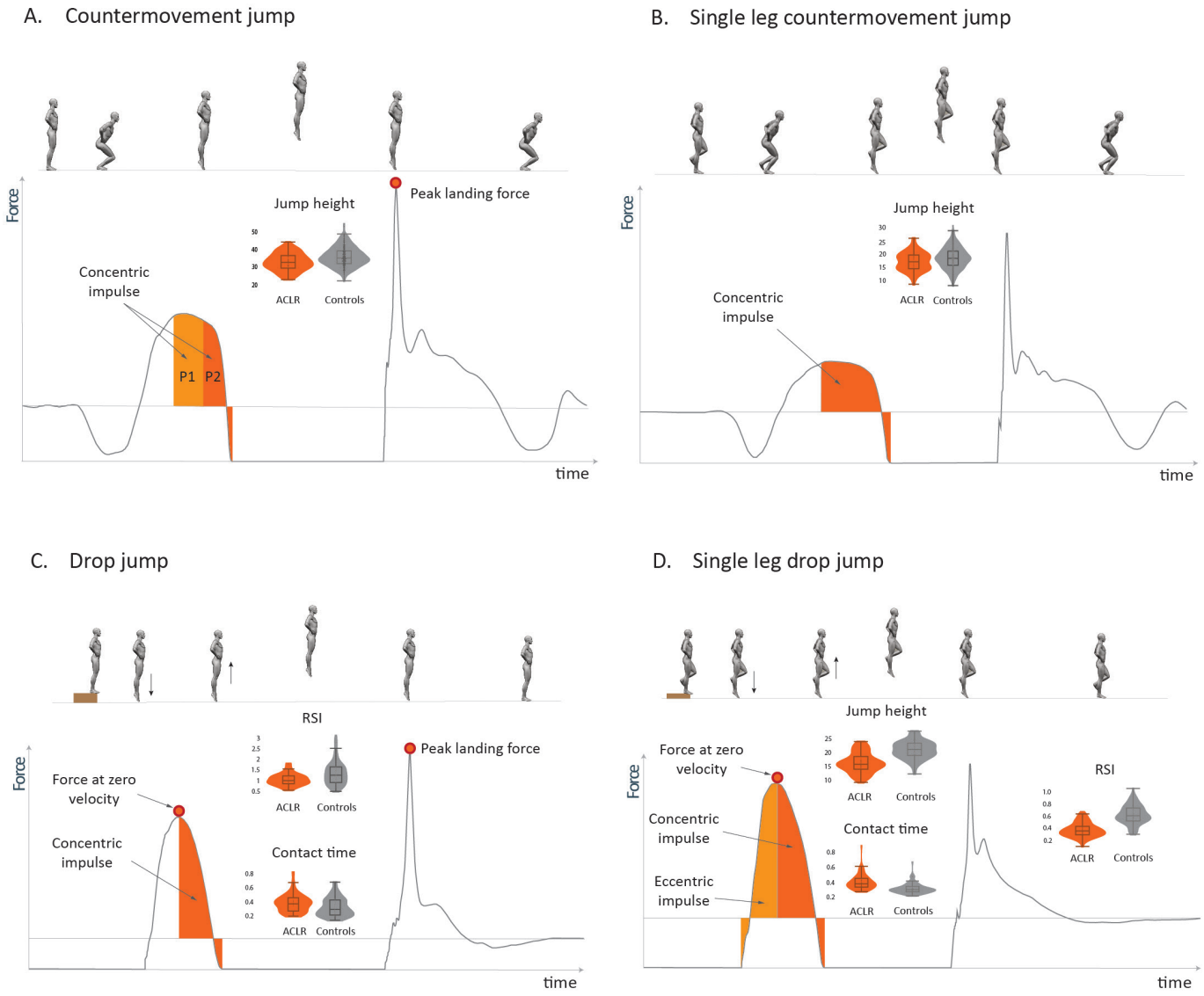

Figure 1 Examples of force-time curves for: (A) countermovement jump, (B) single-leg countermovement jump, (C) drop jump, (D) single-leg drop jump, with the metrics with significant differences between athletes after ACLR and controls. Concentric impulse (N/s): area under the vertical force curve from transition point at zero velocity (max negative displacement) (start of concentric phase) to take-off—net impulse. Concentric impulse P2 (N/s): Net impulse during second 50% (timewise) of concentric phase. Jump height (cm): Jump height calculated by the impulse–momentum method. Peak landing force (N): Peak force as the athlete lands following the jump. Force at zero velocity (N): The force at the instant velocity is zero prior to takeoff. Reactive Strength Index (RSI): Jump height divided by contact time. ACLR, anterior cruciate ligament reconstruction.

Subgroup analysis by graft

Subgroup analysis revealed differences between graft types for the concentric impulse symmetry during the countermovement jump and for force symmetry at zero velocity during the countermovement jumps (single and double) and the single-leg drop jump. Athletes with BTB graft were more asymmetrical than athletes with HS graft. There were no differences between groups for the performance metrics based on graft type (online supplemental file 2).

Performance metrics by activity level and normative values

Performance metrics for the ACLR population (professional and recreational athletes) at the time of RTS, and normative data from professional football players are summarised in table 3.

DISCUSSION

We have identified the vertical jump characteristics which are not yet restored at the time of RTS after ACLR, and we provide normative performance data for professional football players. The most significant residual deficits were in the concentric phase, and in the performance metrics across all jumps investigated. The single-leg drop jump was the most sensitive test to reveal asymmetries in kinetics and performance metrics and the countermovement jump to expose landing asymmetries. The deficits we identify in this large cohort confirm our previous findings: (1) that ACLR patients show significant kinetic deficits despite meeting RTS criteria and (2) that vertical jump tests are sensitive to uncover those deficits. Kinetics asymmetries were most prominent during the concentric phase across all jumps examined. We found no differences in the eccentric deceleration impulse at the time of RTS. The asymmetries still present during the concentric phase are in agreement with previous

Br J Sports Med: first published as 10.1136/bjsports-2022-106588 on 1 June 2023. Downloaded from http://bjsm.bmj.com/ on June 5, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.

Table 2 Comparison of kinetic symmetries between groups during vertical jumps

	Mean±SD	CV%	Mean±SD	CV%	p value	d	Mean difference (95% CIs)
Countermovement jump	**ACLR n=126**		**Controls n=532**				
Eccentric deceleration impulse	96±13	13	97±14	15	0.41		−0.9 (−3.7, 1.8)
Force at zero velocity	96±11	12	97±12	12	0.09		−1.7 (−4.0, 0.6)
Concentric impulse_P1	95±8	9	98±8	8	**<0.001**	−0.42 (−0.61, −0.22)	−3.5 (−5.1, −1.9)
Concentric impulse_P2	90±8	9	100±7	7	**<0.001**	−1.42 (−1.63, −1.21)	−10.3 (−11.7, −8.9)
Concentric impulse	93±7	8	99±7	7	**<0.001**	−0.95 (−1.15, −0.75)	−6.5 (−7.9, −5.2)
Peak landing force	90±20	22	101±19	19	**<0.001**	−0.60 (−0.80, −0.41)	−11.6 (−15.4, −7.9)
Single-leg jump	**ACLR n=118**		**Controls n=516**				
Eccentric deceleration impulse	94±18	19	99±28	28	0.12		−5.0 (−9.0, 0.3)
Force at zero velocity	98±9	10	99±10	10	0.19		−1.4 (−3.4, 0.6)
Concentric impulse	92±6	7	101±8	8	**<0.001**	−1.11 (−1.32, −0.90)	−8.5 (−10.0, −7.0)
Peak landing force	99±12	12	100±13	13	0.91		−0.5 (−3.0, 2.0)
Drop jump	**ACLR n=65**		**Controls n=100**				
Eccentric impulse	92±14	15	98±24	25	0.23		−5.2 (−11.7, 1.2)
Force at zero velocity	92±11	12	97±13	14	**0.036**	−0.35 (−0.67, −0.04)	−4.5 (−8.3, −0.6)
Concentric impulse	89±11	12	100±11	11	**<0.001**	−0.96 (−1.29, −0.63)	−10.5 (−14.0, −7.1)
Peak landing force	93±17	19	101±20	20	**0.023**	−0.46 (−0.78, −0.15)	−8.9 (−14.9, −2.8)
Single-leg drop jump	**ACLR n=110**		**Controls n=123**				
Eccentric impulse	96±7	7	99±8	8	**0.018**	−0.48 (−0.74, −0.22)	−3.8 (−5.8, −1.7)
Force at zero velocity	92±11	12	101±11	11	**<0.001**	−0.84 (−1.11, −0.57)	−9.2 (−12.1, −6.4)
Concentric impulse	92±6	7	101±6	6	**<0.001**	−1.35 (−1.67, −1.05)	−8.8 (−10.4, −7.1)
Peak landing force	98±11	11	99±14	14	0.96		−1.4 (−4.7, 1.8)

Symmetries are presented as percentages ±SD (involved limb/uninvolved limb) for the ACLR cohort and (non-dominant limb/dominant limb) for the control cohort; d: Cohen's d (effect size).
Bold indicates statistically significant differences and their respective effect sizes.
ACLR, anterior cruciate ligament reconstruction.

research in professional football players at 9 months after surgery[29] and previous studies in elite skiers at 2 years after surgery.[30]

Concentric phase asymmetries

During the countermovement jump, the difference in symmetry between groups was more prominent during the second 50% (timewise) of the concentric phase. A possible explanation might be the role of ankle plantar flexors. Our recent research on a similar population showed that soleus muscle contribution was significantly less in both legs of athletes after ACLR compared with a control group during the concentric phase of a horizontal and a vertical jump.[5 7] Previous research suggests

Table 3 Performance metrics by activity level in athletes after ACLR and healthy professional athletes

	Mean±SD	Mean±SD	Mean±SD	Mean±SD	Mean±SD
Countermovement jump	**Recreational athletes—ACLR**	**Professional athletes—ACLR**	**Professional athletes**		
Jump height	28.3±6.0	33.3±5.0	36.3±5.4		
Drop jump	**Recreational athletes—ACLR**	**Professional athletes—ACLR**	**Professional athletes**		
Jump height	27.3±56.7	35.9±5.9	36.1±5.2		
Contact time	0.37±0.09	0.39±0.12	0.34±0.13		
RSI	0.9±0.3	1.09±0.3	1.38±0.5		
Single-leg jump	**Recreational athletes—INV**	**Recreational athletes—UNINV**	**Professional athletes—INV**	**Professional athletes—UNINV**	**Professional athletes—controls**
Jump height	10.5±3.6	12.7±3.4	14.2±3.4	16.5±3.3	17.4±3.7
Single-leg drop jump	**Recreational athletes—INV**	**Recreational athletes—UNINV**	**Professional athletes—INV**	**Professional athletes—UNINV**	**Professional athletes—controls**
Jump height	11.3±3.9	13.7±4.1	14.6±3.5	17.1±3.6	18.5±2.9
Contact time	0.41±0.06	0.38±0.05	0.41±0.10	0.38±0.09	0.33±0.07
RSI	0.29±0.12	0.37±0.14	0.38±0.12	0.47±0.14	0.58±0.13

ACLR, anterior cruciate ligament reconstruction; INV, involved leg; RSI, Reactive Strength Index; UNINV, uninvolved leg.

Br J Sports Med: first published as 10.1136/bjsports-2022-106588 on 1 June 2023. Downloaded from http://bjsm.bmj.com/ on June 5, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.

that medial gastrocnemius onset occurs later in a countermovement jump than other muscles and that peak activation occurs at approximately 83% of the jump duration, which reinforces the suggestion that plantar flexors play an important role in force generation during the late jump phase.[31]

Additionally, a recent study reported a lack of recovery of body velocity in the final phase of take-off and suggested the sustained impairments in plantar flexor muscle function in both the affected and non-affected limbs as a possible reason.[32] Finally, during a single-leg squat jump, a 34% decrease in maximal ankle power between limbs in patients after ACLR was reported, not from an alteration in joint moment but from reduced angular velocity of ankle plantar flexion.[33]

Performance metrics asymmetries

Performance metrics were not fully restored in all types of jumps examined. In unilateral jumps, there are differences between limbs, while in bilateral jumps, there are differences between the ACLR and control groups. Jump performance (jump height) is a critical performance metric that is closely related to the concentric phase of a vertical jump task. The ability to take off with powerful extension of the hip, knee and ankle is essential for achieving maximum jump height.[25] Control participants displayed symmetry in jump height, which suggests that symmetrical measures are critical for assessing rehabilitation completion and should be used as a criterion to ensure that athletes are back to their preinjury levels. It seems that the task of restoring vertical jump symmetry is more challenging than restoring symmetry in strength or hop distance especially since our athletes have exceeded the 90% threshold and were cleared to RTS. Fortunately, for clinicians, kinetic asymmetries are reflected in performance asymmetries, which can be easily detected without the need of advanced equipment. Recent improvements in technology enable the use of valid and reliable alternative methods to measure vertical jump performance, such as low-cost force plates, contact mats, photoelectric cells or even mobile applications.[16–21]

The literature is scarce on performance metrics during vertical jumps in athletes after ACLR. Jump height increases over time following surgery during a countermovement jump[29]; however, at 6 months[34] and 9 months[29] after surgery the jump height remains significantly lower than that of healthy matched controls.

The single-leg drop jump test

The single-leg drop jump showed significant differences between groups in all phases (except landing) and performance outcomes. At the time of RTS after ACLR, athletes still did not achieve symmetry in kinetics and performance metrics, favouring the uninvolved side. Single-leg drop jumps appear to better expose knee deficits in those recovering from ACL injury. Briefly, the individual components of the single-leg drop jump—eccentric and concentric force—shed light on the leg's load acceptance and generating capacity, respectively. Also, highlighted the large differences in performance metrics such as jump height and contact time. Deficits identified in comparison to the uninjured limb and compared with normative data for similar athletes can then be targeted through tailored rehabilitation. Reactive tests (drop jumps) are important to evaluate other performance parameters such as power and reactivity. Plyometric training may improve subjective function and functional activities compared with usual rehabilitation care, without any increase in laxity or pain and should be an integral part of the rehabilitation protocol.[35] Most research on reactive jumps emphasises double-leg drop jumps[36–38] and there is not much research on single-leg drop jump metrics after ACLR.[39] Our recent study reported a 70% difference in knee work between limbs in the concentric phase in athletes at the time of RTS.[7] Residual deficits during single-leg drop jump were evident also in the performance metrics with asymmetries of 20%–30% for jump height and RSI at 9 months after surgery.[7 9]

Bilateral or unilateral test?

Bilateral and unilateral tests are both important and provide different information for the athlete's status. Bilateral tests are useful to evaluate a patient's interlimb compensatory movement strategies and unilateral tests offer an opportunity to assess the performance of each limb capacity in isolation which can additionally be compared with the contralateral limb as well as normative reference data. In the current study, we found differences in both double-leg tasks (countermovement and drop jump) during landing, indicating that athletes still shift their weight during landing to the uninvolved leg. In our exploratory analysis during the countermovement jump, we evaluated results for both peak landing force and landing impulse. The landing impulse metric had greater variance (trial-to-trial differences) in both athletes after ACLR and controls, so we suggest using the peak landing force metric instead of landing impulse. Unilateral tests are useful to evaluate the ability of each limb to perform independently. In patients after ACLR it is common to compensate for lower knee work with greater hip and ankle work,[5 7] however, these intralimb compensations cannot be detected by the uniaxial force plates and require 3D biomechanical analysis, apparatus not frequently available in the clinical setting. In the absence of advanced equipment, clinicians can use performance metrics such as jump height—during unilateral vertical tests to evaluate the overall lower limb functional status of the athletes. Peak landing force in unilateral tests did not show statistically or clinically important differences and is not recommended as a useful metric.

Influence of graft choice

Graft choice had an influence on kinetic symmetry in specific metrics, with BTB procedures resulting in greater asymmetries. The transition between eccentric and concentric phase (force at zero velocity) was the metric influenced mostly by the graft type, with greater asymmetries with a BTB graft in all tests. Previous research reports that during a countermovement jump at 9 months after ACLR, athletes with a BTB graft had greater inter-limb impulse asymmetries than athletes with a HS graft in the eccentric deceleration and concentric phases of the countermovement jump, although similar jump heights were achieved.[40] During landing there was a significant difference between ACLR and control groups, but no difference between BTB and HS graft, in agreement with previous research.[40]

Is symmetry important?

The goal of rehabilitation is to return the athlete back to normal. It is difficult to define normal, as this is different for each patient. Clinicians should use asymmetry metrics on an individual level and by comparing to the "noise" of each test and each metric.[41] Normally, the clinician does not have preoperative test values to set the end goals for each patient.[42] Achieving symmetry is an important goal during rehabilitation, but equally important is to return the athlete to their previous level of performance. We suggest that the uninvolved limb should be monitored during

Br J Sports Med: first published as 10.1136/bjsports-2022-106588 on 1 June 2023. Downloaded from http://bjsm.bmj.com/ on June 5, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.

rehabilitation, and both limbs should reach matched-control normative values in the absence of preoperative data. In our healthy cohort, we report the performance outcomes for each of the tests. These can be used by the clinicians to set a target for their patients where preoperative values are not available. Although symmetry and its importance has been questioned, we cannot deny the fact that, consistent with previous findings,[30 32] the between-limb asymmetry in the control group was small in all the variables explored, except landing. The SD of each variable is also reported to help the clinician interpret any differences. There was no statistically significant difference between limbs in the control group for performance metrics during the single-leg jumps. Limb dominance did not affect the performance of these tests in football players; this might not apply in other sports.

Clinical implications

We present between-leg asymmetries during four vertical jump tests in athletes after ACLR who were cleared to RTS and compare them with a large healthy cohort. Clinicians can use these results to inform their clinical practice and restore the phase-specific asymmetries. We emphasise the importance of restoring symmetry between legs—not only in jump testing—in patients who undergo a long rehabilitation period; however, this is not enough. Absolute performance metrics are not fully restored for the involved and the uninvolved leg and clinicians should target also restoring vertical jump capacity in both legs. Our study provides performance metrics for the ACLR population (professional and recreational athletes) at the time of RTS, and normative data from professional football players which can be used as a target by clinicians working with similar populations.

The residual deficits seen in single-leg drop jump suggest that athletes were not exposed enough in plyometric training. Plyometric drills are fundamental components of most sports and an important neuromuscular quality closely associated with performance in explosive sports.[43] Additionally, rehabilitation should target not only strength restoration of the quadriceps and hamstrings, but also the ankle plantar flexors as they might be an important component of vertical jump capacity.

Bilateral jumps best described landing asymmetries, but clinicians should be mindful of the relatively high population variance, even in the control group, when interpreting the results.

Limitations and future directions

We included only males from a single institution and the generalisability of these results to other populations is unknown. Particularly the performance normative metrics and the asymmetry results (with their corresponding variance) can vary widely based on sport, activity level and gender which limits comparisons between groups. In the current study, normative data are for professional football players and the results presented might be different in other sports. We did not include normative data for recreational athletes to evaluate if our patients restored performance metrics or not. Future studies should report vertical jump performance metrics for female and recreational athletes of different sports. We investigated outcomes at a single time point, at the time of RTS. During rehabilitation, there might be additional important metrics to track an athlete's progression that normalise before the time of RTS. We need longitudinal studies during the rehabilitation period to describe these metrics. Correct execution of the tests is an important parameter that can affect the results. Although we instructed athletes to countermove fast, we did not control the peak eccentric velocity, that is, we did not set a threshold on that metric to increase consistency. However, exploring the results, there was no difference in peak eccentric velocity between groups.

All metrics that were examined have been presented, yet these objectively large number of measures remain a subset of all possible jump metrics. Future research should consider preregistration of outcomes to reduce the possibility of false positive results. More studies are needed to evaluate the differences found mostly in the concentric phase and if this finding is related to the rehabilitation approach or it is a common finding seen in athletes after ACLR. Further research is required to determine the value of vertical jump testing at the time of RTS and the relationship between the residual functional asymmetries and the risk of future injury (eg, graft tear or contralateral ACL injury, meniscal and chondral injuries, osteoarthritis).

Conclusion

Vertical jumps testing offers valuable phase-specific information on the status of the athlete after ACLR and should be included in the periodic and RTS assessment. In our cohort, there were still asymmetries in the concentric (push-off) phase, during landing of bilateral jumps and most of the performance metrics, despite passing the traditional discharge criteria. Apart from symmetry clinicians should target restoring absolute performance metrics such as jump height, RSI and contact times.

Twitter Roula Kotsifaki @RoulaKotsifaki, Vasileios Sideris @vasilisbme, Enda King @enda-king, Roald Bahr @RoaldBahr and Rod Whiteley @RodWhiteley

Acknowledgements The authors thank the physiotherapists (ACL Group) of Aspetar Rehabilitation Department for assisting with subjects' recruitment.

Contributors RK and RW participated in the conception and the design of the study. RK and VS were responsible for data collection. RK performed the data analysis and table designs and all the authors contributed to the interpretation. RK drafted the manuscript, and all the authors revised it critically and gave their approval of the final version. RK acted as the guarantor for the study.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants and was approved by Aspetar Institutional Review Board. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available on reasonable request.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

ORCID iDs
Roula Kotsifaki http://orcid.org/0000-0002-7902-9206
Enda King http://orcid.org/0000-0003-0434-1489
Roald Bahr http://orcid.org/0000-0001-5725-4237
Rod Whiteley http://orcid.org/0000-0002-1452-6228

REFERENCES

1 Kyritsis P, Bahr R, Landreau P, *et al*. Likelihood of ACL graft rupture: Not meeting six clinical discharge criteria before return to sport is associated with a four times greater risk of rupture. *Br J Sports Med* 2016;50:946–51.

Br J Sports Med: first published as 10.1136/bjsports-2022-106588 on 1 June 2023. Downloaded from http://bjsm.bmj.com/ on June 5, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.

2 Abrams GD, Harris JD, Gupta AK, *et al*. Functional performance testing after anterior Cruciate ligament reconstruction: A systematic review. *Orthop J Sports Med* 2014;2:2325967113518305.
3 Kotsifaki A, Korakakis V, Whiteley R, *et al*. Measuring only hop distance during single leg hop testing is insufficient to detect deficits in knee function after ACL reconstruction: A systematic review and meta-analysis. *Br J Sports Med* 2020;54:139–53.
4 Kotsifaki A, Van Rossom S, Whiteley R, *et al*. Symmetry in triple hop distance hides Asymmetries in knee function after ACL reconstruction in athletes at return to sports. *Am J Sports Med* 2022;50:441–50.
5 Kotsifaki A, Whiteley R, Van Rossom S, *et al*. Single leg hop for distance symmetry masks lower limb Biomechanics: Time to discuss hop distance as decision criterion for return to sport after ACL reconstruction *Br J Sports Med* 2022;56:249–56.
6 Kotsifaki A, Korakakis V, Graham-Smith P, *et al*. Vertical and horizontal hop performance: Contributions of the hip, knee, and ankle. *Sports Health* 2021;13:128–35.
7 Kotsifaki A, Van Rossom S, Whiteley R, *et al*. Single leg vertical jump performance identifies knee function deficits at return to sport after ACL reconstruction in male athletes. *Br J Sports Med* 2022;56:490–8.
8 Ebert JR, Du Preez L, Furzer B, *et al*. Which hop tests can best identify functional limb asymmetry in patients 9-12 months after anterior Cruciate ligament reconstruction employing a hamstrings tendon Autograft *Int J Sports Phys Ther* 2021;16:393–403.
9 King E, Richter C, Franklyn-Miller A, *et al*. Whole-body Biomechanical differences between limbs exist 9 months after ACL reconstruction across jump/landing tasks. *Scand J Med Sci Sports* 2018;28:2567–78.
10 King E, Richter C, Franklyn-Miller A, *et al*. Back to normal symmetry? Biomechanical variables remain more Asymmetrical than normal during jump and change-of-direction testing 9 months after anterior Cruciate ligament reconstruction. *Am J Sports Med* 2019;47:1175–85.
11 Ohji S, Aizawa J, Hirohata K, *et al*. Single-leg hop can result in higher limb symmetry index than Isokinetic strength and single-leg vertical jump following anterior Cruciate ligament reconstruction. *Knee* 2021;29:160–6.
12 Bishop C, Jordan M, Torres-Ronda L, *et al*. Selecting Metrics that matter: Comparing the use of the Countermovement jump for performance profiling, neuromuscular fatigue monitoring, and injury rehabilitation testing. *Strength Cond J* 2023:10.
13 Bishop C, Turner A, Jarvis P, *et al*. Considerations for selecting field-based strength and power fitness tests to measure Asymmetries. *J Strength Cond Res* 2017;31:2635–44.
14 Bishop C, Turner A, Jordan M, *et al*. A framework to guide practitioners for selecting Metrics during the Countermovement and drop jump tests. *Strength Cond J* 2022;44:95–103.
15 Cohen D, Burton A, Wells C, *et al*. Single vs double leg Countermovement jump tests: Not half an apple! *Aspetar Sport Med J* 2020;9:34–41.
16 Brooks ER, Benson AC, Bruce LM. Novel Technologies found to be valid and reliable for the measurement of vertical jump height with jump-and-reach testing. *J Strength Cond Res* 2018;32:2838–45.
17 Haynes T, Bishop C, Antrobus M, *et al*. The validity and reliability of the my jump 2 App for measuring the reactive strength index and drop jump performance. *J Sports Med Phys Fitness* 2019;59:253–8.
18 Healy R, Kenny IC, Harrison AJ. Assessing reactive strength measures in jumping and hopping using the Optojump™ system . *J Hum Kinet* 2016;54:23–32.
19 Pueo B, Lipinska P, Jiménez-Olmedo JM, *et al*. Accuracy of jump-mat systems for measuring jump height. *Int J Sports Physiol Perform* 2017;12:959–63.
20 Balsalobre-Fernández C, Glaister M, Lockey RA. The validity and reliability of an iPhone App for measuring vertical jump performance. *J Sports Sci* 2015;33:1574–9.
21 Stanton R, Kean CO, Scanlan AT. My jump for vertical jump assessment. *Br J Sports Med* 2015;49:1157–8.
22 Ardern CL, Glasgow P, Schneiders A, *et al*. Consensus statement on return to sport from the first world Congress in sports physical therapy, Bern. *Br J Sports Med* 2016;50:853–64.
23 van Melick N, Meddeler BM, Hoogeboom TJ, *et al*. How to determine leg dominance: The agreement between self-reported and observed performance in healthy adults. *PLoS One* 2017;12:e0189876.
24 Healy R, Kenny IC, Harrison AJ. Reactive strength index: A poor indicator of reactive strength *Int J Sports Physiol Perform* 2018;13:802–9.
25 Linthorne NP. Analysis of standing vertical jumps using a force platform. *American Journal of Physics* 2001;69:1198–204.
26 Noyes FR, Barber SD, Mangine RE. Abnormal lower limb symmetry determined by function hop tests after anterior Cruciate ligament rupture. *Am J Sports Med* 1991;19:513–8.
27 Shapiro SS, Wilk MB. An analysis of variance test for Normality (complete samples). *Biometrika* 1965;52(3/4):591.
28 Hedges L, Olkin I. *Statistical methods for meta-analysis*. Academic press, 2014.
29 Read PJ, Michael Auliffe S, Wilson MG, *et al*. Lower limb kinetic Asymmetries in professional Soccer players with and without anterior Cruciate ligament reconstruction: Nine months is not enough time to restore "functional" symmetry or return to performance. *Am J Sports Med* 2020;48:1365–73.
30 Jordan MJ, Aagaard P, Herzog W. Lower limb asymmetry in mechanical muscle function: A comparison between ski racers with and without ACL reconstruction. *Scand J Med Sci Sports* 2015;25:e301–9.
31 Sahrom SB, Wilkie JC, Nosaka K, *et al*. The use of Yank-time signal as an alternative to identify Kinematic events and define phases in human Countermovement jumping. *R Soc Open Sci* 2020;7:192093.
32 Jordan MJ, Morris N, Nimphius S, *et al*. Attenuated lower limb stretch-shorten-cycle capacity in ACL injured vs. non-injured female Alpine ski racers: Not just a matter of between-limb asymmetry. *Front Sports Act Living* 2022;4:853701.
33 de Fontenay BP, Argaud S, Blache Y, *et al*. Motion alterations after anterior Cruciate ligament reconstruction: Comparison of the injured and Uninjured lower limbs during a single-legged jump. *J Athl Train* 2014;49:311–6.
34 O'Malley E, Richter C, King E, *et al*. Countermovement jump and Isokinetic Dynamometry as measures of rehabilitation status after anterior Cruciate ligament reconstruction. *J Athl Train* 2018;53:687–95.
35 Kotsifaki R, Korakakis V, King E, *et al*. Aspetar clinical practice guideline on rehabilitation after anterior Cruciate ligament reconstruction. *Br J Sports Med* 2023;57:500–14.
36 Myer GD, Ford KR, Khoury J, *et al*. Development and validation of a clinic-based prediction tool to identify female athletes at high risk for anterior Cruciate ligament injury. *Am J Sports Med* 2010;38:2025–33.
37 Nilstad A, Andersen TE, Kristianslund E, *et al*. Physiotherapists can identify female football players with high knee Valgus angles during vertical drop jumps using real-time observational screening. *J Orthop Sports Phys Ther* 2014;44:358–65.
38 Noyes FR, Barber-Westin SD, Fleckenstein C, *et al*. The drop-jump screening test: Difference in lower limb control by gender and effect of neuromuscular training in female athletes. *Am J Sports Med* 2005;33:197–207.
39 Maestroni L, Read P, Turner A, *et al*. Strength, rate of force development, power and reactive strength in adult male athletic populations post anterior Cruciate ligament reconstruction - A systematic review and meta-analysis. *Physical Therapy in Sport* 2021;47:91–104.
40 Miles JJ, King E, Falvey ÉC, *et al*. Patellar and Hamstring Autografts are associated with different jump task loading Asymmetries after ACL reconstruction. *Scand J Med Sci Sports* 2019;29:1212–22.
41 Bishop C, Lake J, Loturco I, *et al*. Interlimb Asymmetries: The need for an individual approach to data analysis. *J Strength Cond Res* 2021;35:695–701.
42 Wellsandt E, Failla MJ, Snyder-Mackler L. Limb symmetry indexes can overestimate knee function after anterior Cruciate ligament injury. *J Orthop Sports Phys Ther* 2017;47:334–8.
43 Booth MA, Orr R. Effects of Plyometric training on sports performance. *Strength Cond J* 2016;38:30–7.

Br J Sports Med: first published as 10.1136/bjsports-2022-106588 on 1 June 2023. Downloaded from http://bjsm.bmj.com/ on June 5, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.